

Projet n°027

Titre : A CAUSAL APPROACH TO BITCOIN PERFORMANCE MODELING

Description : Bitcoin journey has not been a long calm river since the Genesis Block on Jan. 3, 2009. It went through major ups and downs, came under countless technical/legal/moral attacks. Yet, 15+ years later, network activity is hovering near historical highs, and institutional investors are assembling at the gate. Against all odds, bitcoin is indeed progressively becoming eligible to a larger crowd of investors, as the infrastructure stack is getting more robust, the legal framework is being drawn up, and new investment solutions are coming to the market. There is still one missing piece though for the king of crypto assets to reach the tipping point, and see a massive arrival of institutional investors: a better understanding of the key determinants of its performance. The primary objective of this project is, in this respect, to develop a causal model and provide institutional investors with robust tools enabling them to make sense of the behavior of bitcoin. A secondary objective of this project will be to build on these models to design a slow-moving investment strategy aiming at capturing the upside potential while mitigating the downside risk.

Brute force yields great out-of-sample results when data is abundant, and the observed phenomenon is stable. It is a different story when data is scarce (i.e., only 15 years of track record in the case of bitcoin), and the phenomenon under scrutiny is evolving (i.e., the behaviors of the miners/market makers/hedgers/investors/speculators is adapting to a changing environment). Not to mention the fact that an institutional investor is an Agent, and as such, he must be able to rationalize his investment decisions, and afterwards, to explain in plain English the performance of his strategy, to the Principal; which is far from trivial if the decision was made with sophisticated machine learning algorithms. Hence the idea to try and build a causal model to explain the performance of bitcoin. We will strive to answer two simple questions: Is it possible to build causal models that could help investors forecast bitcoin's next day/week/month/quarter/year performance? If so, to what extent do the models change with the investment horizon?

Deliverable: Jupyter Notebook covering the whole process (data pipeline, feature selection, causal graph, tests of accuracy, comparison with machine learning algorithms, etc.)

Informations sur le projet

Compétences développées :

Informations complémentaires :

Majeure(s) concernée(s) : DIA, IF, FIN

Année(s) concernée(s) : A4, A5

Acceptera une équipe A4 si aucune équipe A5 ne se positionne : Oui

Mot(s)-clé(s) concerné(s) : Blockchain, Cryptomonnaies, Data Science, DataMining, Finance quantitative, Gestion d'actifs, Machine learning, Programmation, Python

Informations sur le partenaire

Entreprise / Association / École : GINJER

Nom : VAISSIE

Prénom : MATHIEU